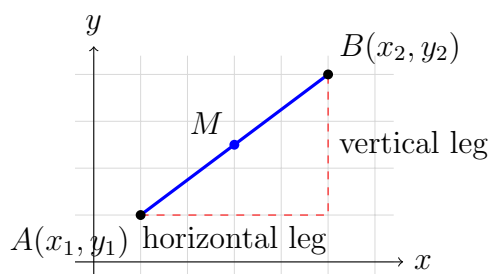


# Computing Distance and Midpoint from Coordinates

**Where the two formulas come from.** Drop a horizontal leg and a vertical leg from the segment's endpoints. The segment is the hypotenuse of a right triangle whose legs are  $|x_2 - x_1|$  and  $|y_2 - y_1|$ , so the Pythagorean theorem gives the length. The midpoint is not a length at all — it is the *average* of the two endpoints, taken one coordinate at a time.



How every segment on this sheet is labelled:  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are the endpoints and  $M$  is the midpoint. No values are shown here — use the coordinates each problem gives you.

- Distance:  $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint:  $M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- Leave a distance in exact radical form unless the problem says otherwise; show the substitution step before you simplify.

**1.** Find the distance between  $A(1, 2)$  and  $B(4, 6)$ . Show the substitution into the distance formula, and name the two legs of the right triangle you are using.

Answer: \_\_\_\_\_

- 2.** Find the midpoint of the segment joining  $A(2, -3)$  and  $B(8, 7)$ . Write it as an ordered pair.

Answer: \_\_\_\_\_

- 3.** Find the exact distance between  $A(-3, 1)$  and  $B(2, 13)$ . Watch the signs when you subtract the negative  $x$ -coordinate.

Answer: \_\_\_\_\_

- 4.** Find the midpoint of  $A(-5, 4)$  and  $B(3, -2)$ . Then state which quadrant the midpoint lies in.

Answer: \_\_\_\_\_

**5.** Find the distance between the origin  $A(0,0)$  and  $B(6,6)$ . Give the answer in exact radical form, fully simplified — not a rounded decimal.

Answer: \_\_\_\_\_

**6.**  $M(3,5)$  is the midpoint of  $\overline{AB}$ , and one endpoint is  $A(-1,2)$ . Find the coordinates of  $B$ .

(Think of the midpoint formula as two averaging equations, one for  $x$  and one for  $y$ , and solve each for the missing coordinate. Then check your  $B$  by recomputing the midpoint.)

Answer: \_\_\_\_\_

**7.** Triangle  $PQR$  has vertices  $P(1,1)$ ,  $Q(7,1)$ , and  $R(4,6)$ .

(a) Find the midpoint  $M$  of side  $\overline{PQ}$ .

Answer: \_\_\_\_\_

(b) Find the length of the median  $\overline{RM}$  — the segment from  $R$  to that midpoint.

Answer: \_\_\_\_\_

**8. Coordinate proof.** Quadrilateral  $ABCD$  has vertices  $A(0, 0)$ ,  $B(6, 2)$ ,  $C(8, 7)$ , and  $D(2, 5)$ . Its diagonals are  $\overline{AC}$  and  $\overline{BD}$ .

Compute the midpoint of each diagonal. Then write a short proof: state what you found and explain why sharing a midpoint proves that the diagonals **bisect each other**. Two or three complete sentences.